Monthly Land Surface Temperature (LST) and Urban Heat Island (UHI) Dataset for Birmingham from Jan 2025 to Dec 2025

The dataset contains monthly LST and monthly UHI raster's derived from LST observations from the Moderate Resolution Imaging Spectroradiometer (MODIS) Land Surface Temperature and Emissivity Daily L3 Global 1 km product (MOD11A1), which provides daytime surface temperature at 1 km resolution. The monthly LST is obtained by calculating the average of all the LST values available at that month on each pixel. UHI is calculated as the difference between the LST of each pixel and the mean LST of the region. Positive values represent areas warmer than average, and negative values represent relatively cooler areas.

Study Area

The study area includes the city of Birmingham and the surrounding region in Shelby and Jefferson counties. Bounding coordinates of the datasets are:

Direction	Coordinates
West	-87.343195
East	-86.337082
South	33.013087
North	33.848520

Data Source

Satellite instrument: MODIS

Primary satellite product: MOD11A1 (this product provides global LST at 1km spatial resolution) (https://ladsweb.modaps.eosdis.nasa.gov/)

Temporal Coverage

Temporal resolution: monthly

Data type: monthly rasters for UHI and LST

Time period: 2025 Jan to 2025 Dec

Each raster represents the average of all available MODIS LST observations for that month.

Example file naming convention:

{year}_{Month}_LST.tif

Eg: 2025_01_LST.tif-------LST raster for January of 2025 →contains value in °C

{year}_{Month}_UHI.tif

Eg: 2025_01_UHI.tif ------UHI raster for January of 2025 →contain value in °C

Data Processing methodology

- Download daily LST images from MODIS for a particular month.
- Clip the LST images to the Birmingham study area boundary.
- Compute the monthly mean LST using all available observations within each month.
- Calculate the mean LST across the entire region
- Compute the Urban Heat Island intensity for each pixel using:

$$UHI = LST_{pixel} - LST_{regional\ mean}$$

- Export resulting UHI layers as a Geotiff raster file.

Raster information

Columns	112
Rows	93
Number of Bands	1
Cell Size X	0.008983152841195188
Cell Size Y	0.008983152841195182
Uncompressed Size	5.09 KB
Format	TIFF
Source Type	Generic
Pixel Type	unsigned char
Pixel Depth	4 Bit
NoData Value	15
Colormap	absent
Pyramids	absent
Compression	LZW
Mensuration Capabilities	Basic

Interpretation of values for Urban Heat raster

Value Range	Interpretation
Positive values	Warmer than regional average (urban heat island effect)
Negative values	Cooler than regional average
Values near zero	Similar to regional mean temperature

Data Limitation

- Cloud contamination may reduce the number of valid observations in some months.
- The fine-scale urban thermal variability is not captured by 1 km resolution.